

Self-distillation of context — cross-checkout report

Self-distillation of context – cross-checkout comparison.

Method families compared (one row per run, best-effort artifacts):

[layerwise] 159 runs

Forward layerwise distillation (this checkout). Every block updated with uniform k-deep credit; behavioral readout teacher-sourced.

[classical-kd] 5 runs

Classical/naive KD baselines (../selfupdate_kd): full fine-tune and LoRA KL-distillation of the top layers.

[multigpu] 18 runs

Big-model layerwise on sharded GPUs (../selfupdate_multi): gpt-oss-120B, Qwen3.5-122B, Gemma-4, ALIA-40B.

Metric switch (owner directive, 2026-07-07): the tiny held-out general cross-entropy (log-loss) 'destruction' canary was measured to be noise-dominated and near-flat across training. It is replaced by two standard, bounded, low-variance measurements, scored in a single teacher-forced batched pass so only model LOADING costs time:

DESTRUCTION (capability retained vs the epoch-0 teacher of the base):

- ARC-Easy accuracy on a fixed cached subset (option log-likelihood)
- WikiText-2 validation perplexity (quantization-style damage)

RECALL (memorization of the original text; eval-only, never trained):

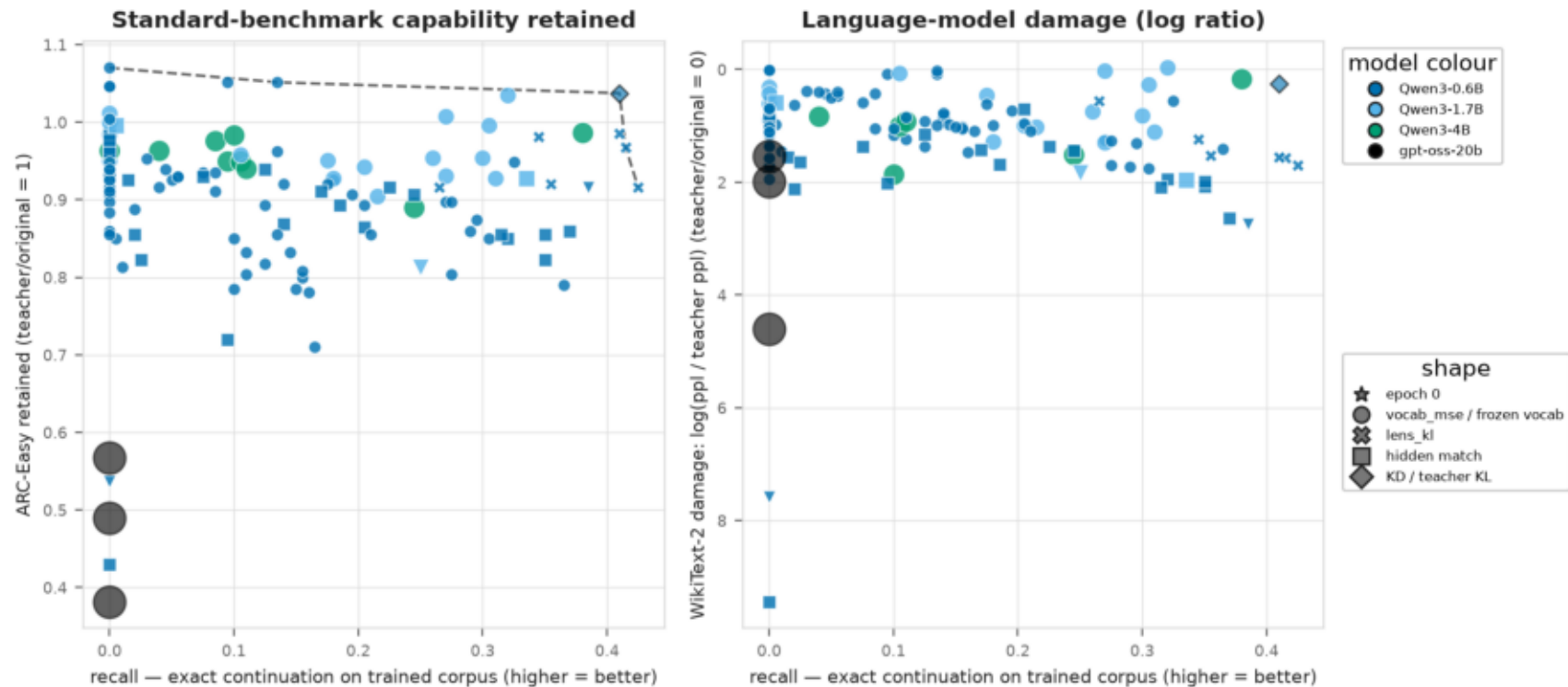
- exact continuation + interior-word cloze (Machado verse)
- next-sentence continuation + multi-word paragraph censor (Cervantes)

Coverage: 171 trained runs total; 129 with the new retention battery, 148 with recitation CER. The index also includes 11 epoch-0 teacher/original baseline rows, where ARC retained = 1 and WikiText log perplexity-ratio damage = 0. Runs still needing the battery are listed on the coverage page; rerun scripts/retention_eval.py to fill them.

Generated 2026-07-08 12:25. Backbone: runs/retention_index.csv.

Recall vs capability retention (bidimensional)

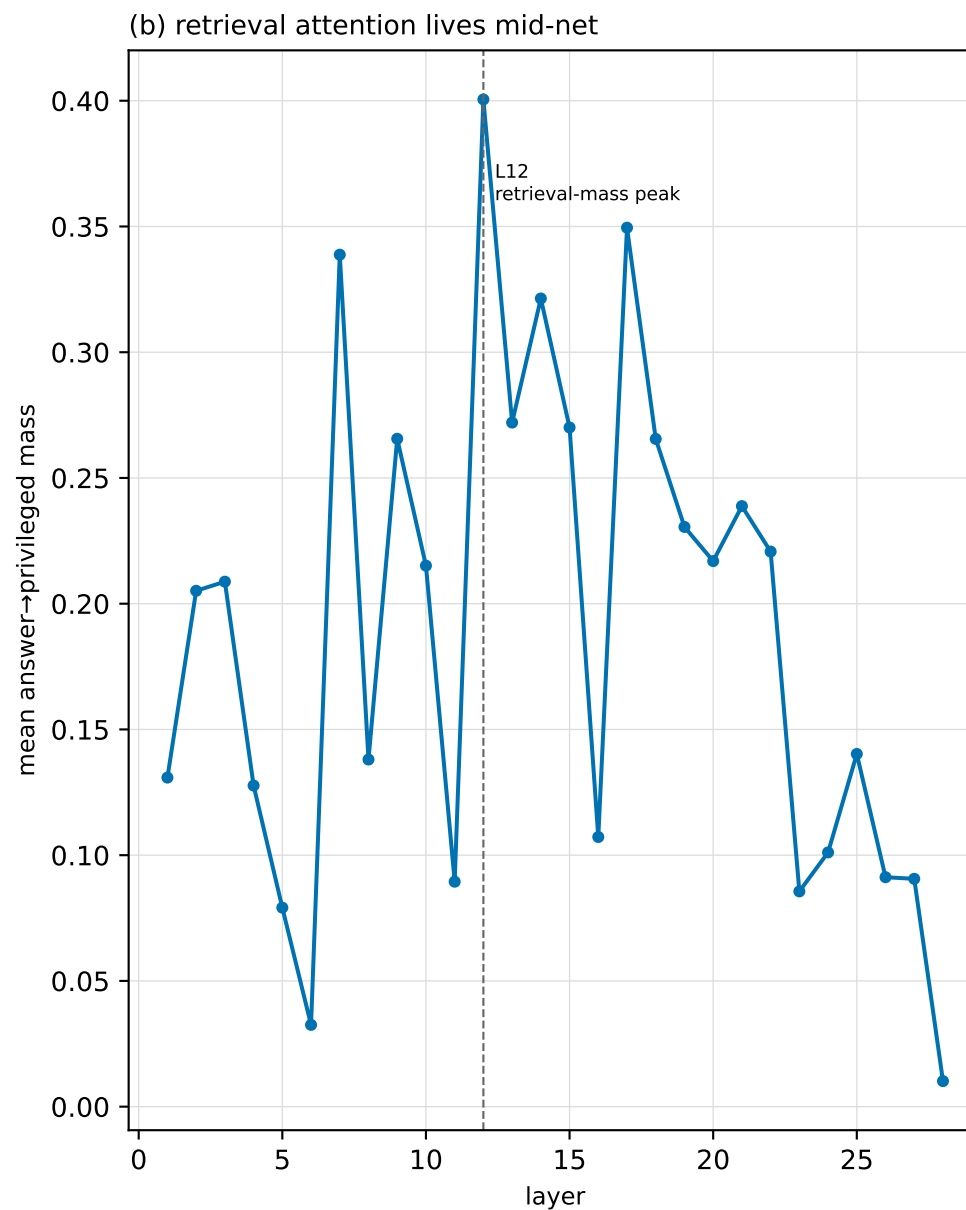
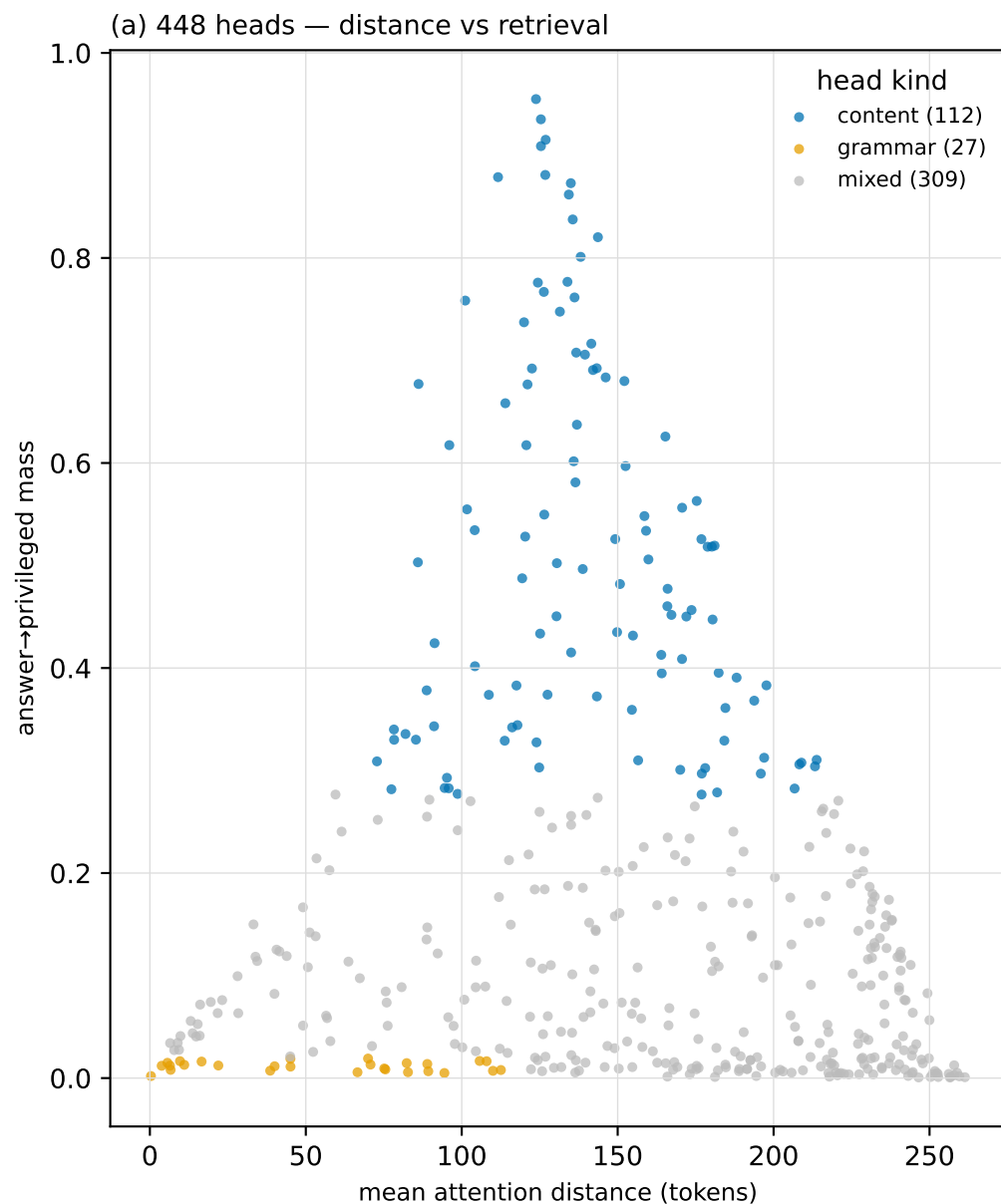
Recall vs capability retention — endpoint Pareto across method families



129 rows with retention evidence, including epoch-0 teacher anchors · colour = model · shape = loss/lens/readout kind · marker size ~ model params

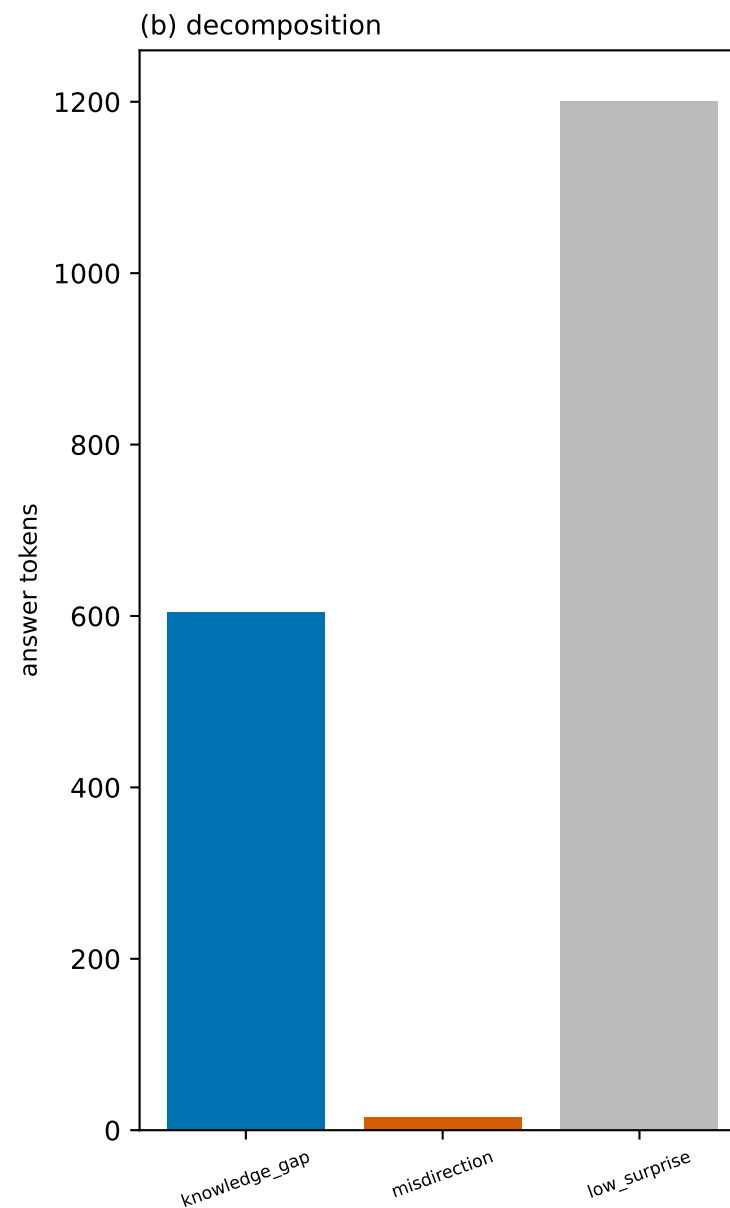
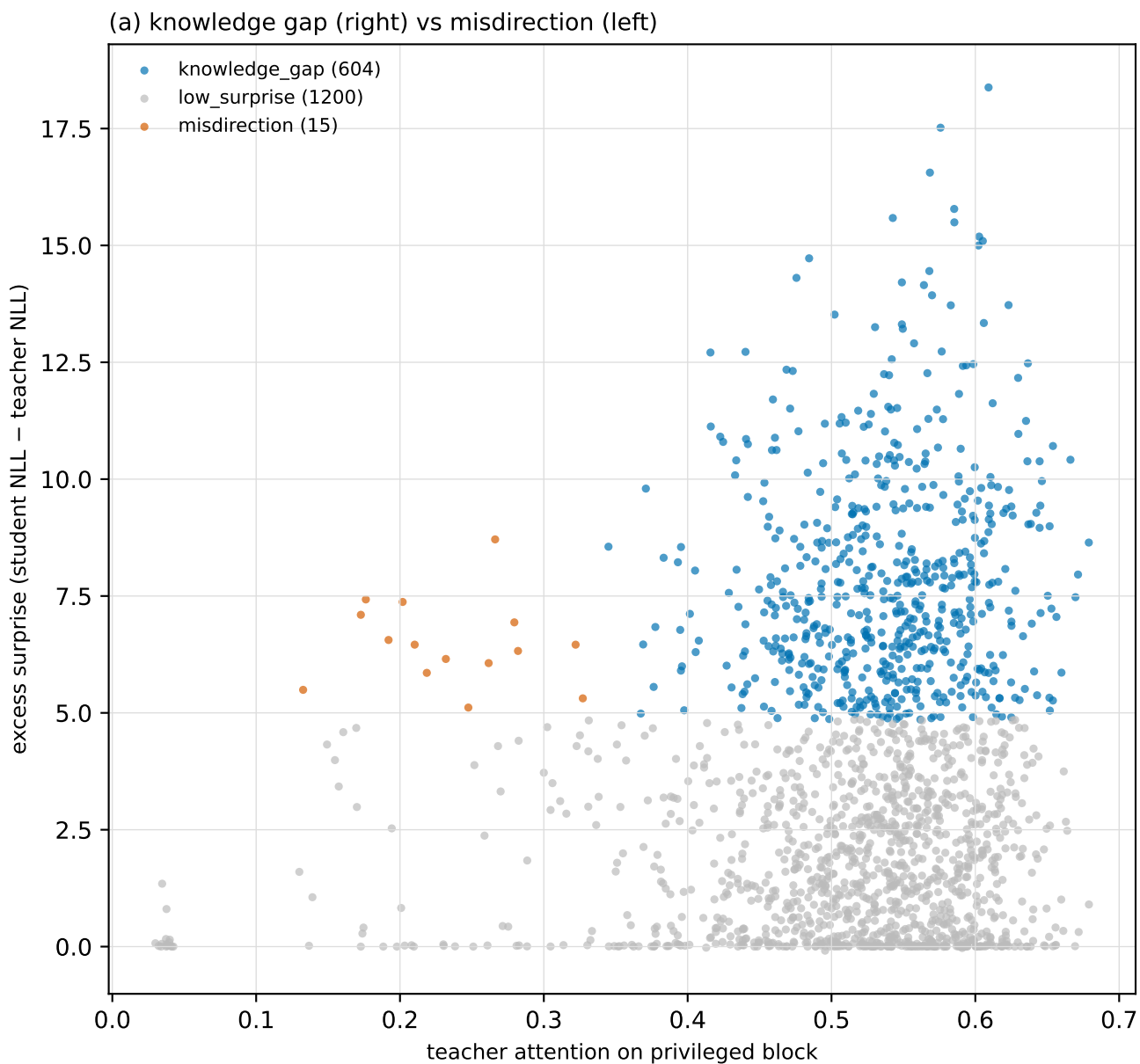
What to memorize — attention head taxonomy (0.6B)

Teacher view (privileged passage present), answer positions. A head that reaches FAR (high distance) and lands ON the privileged block (high answer→privileged mass) is a content/insight head — it marks what the context considered worth absorbing. Local, privileged-blind heads are grammar. This is the memorization target that motivates hidden-primary (mid-net) storage, and it names what the recall probes should test.



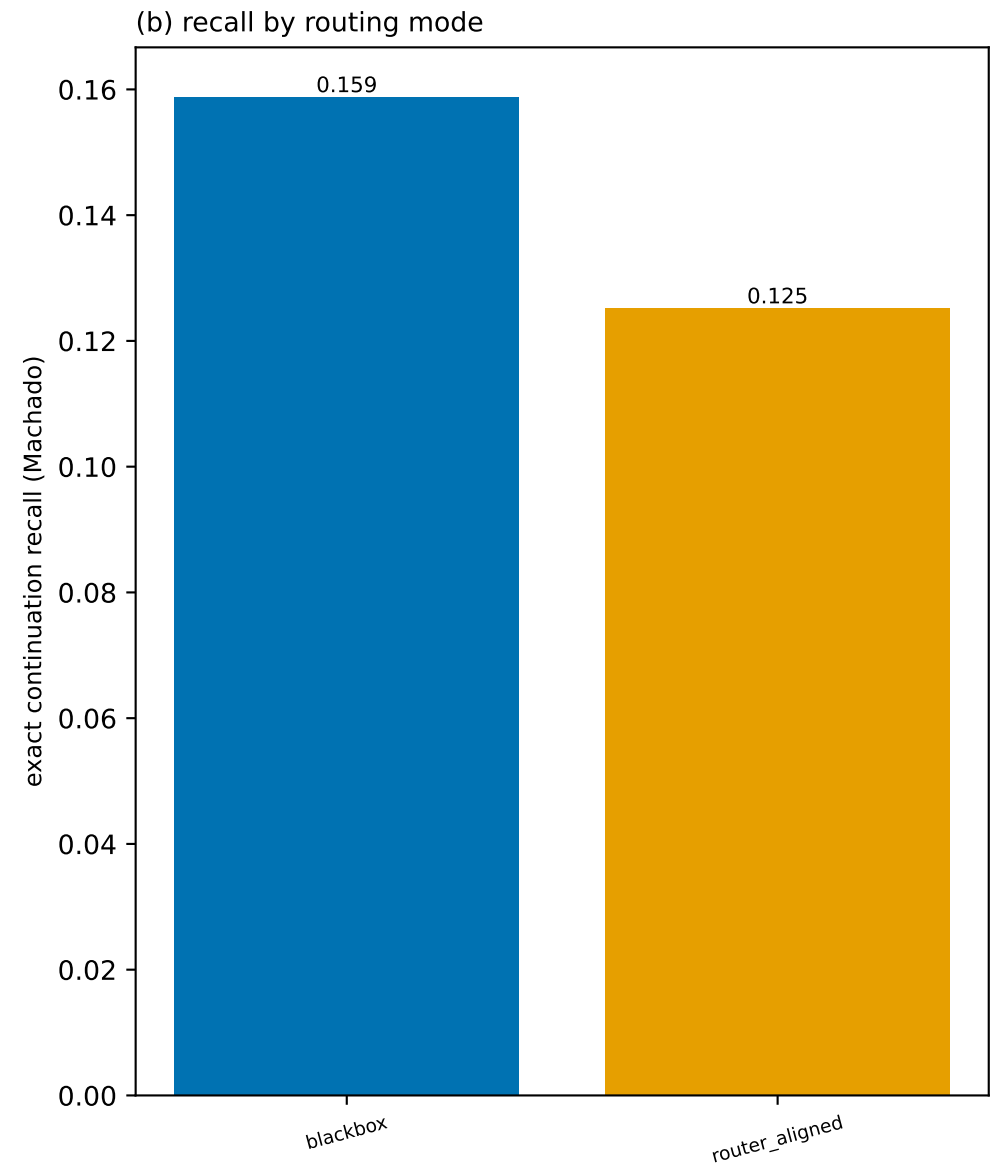
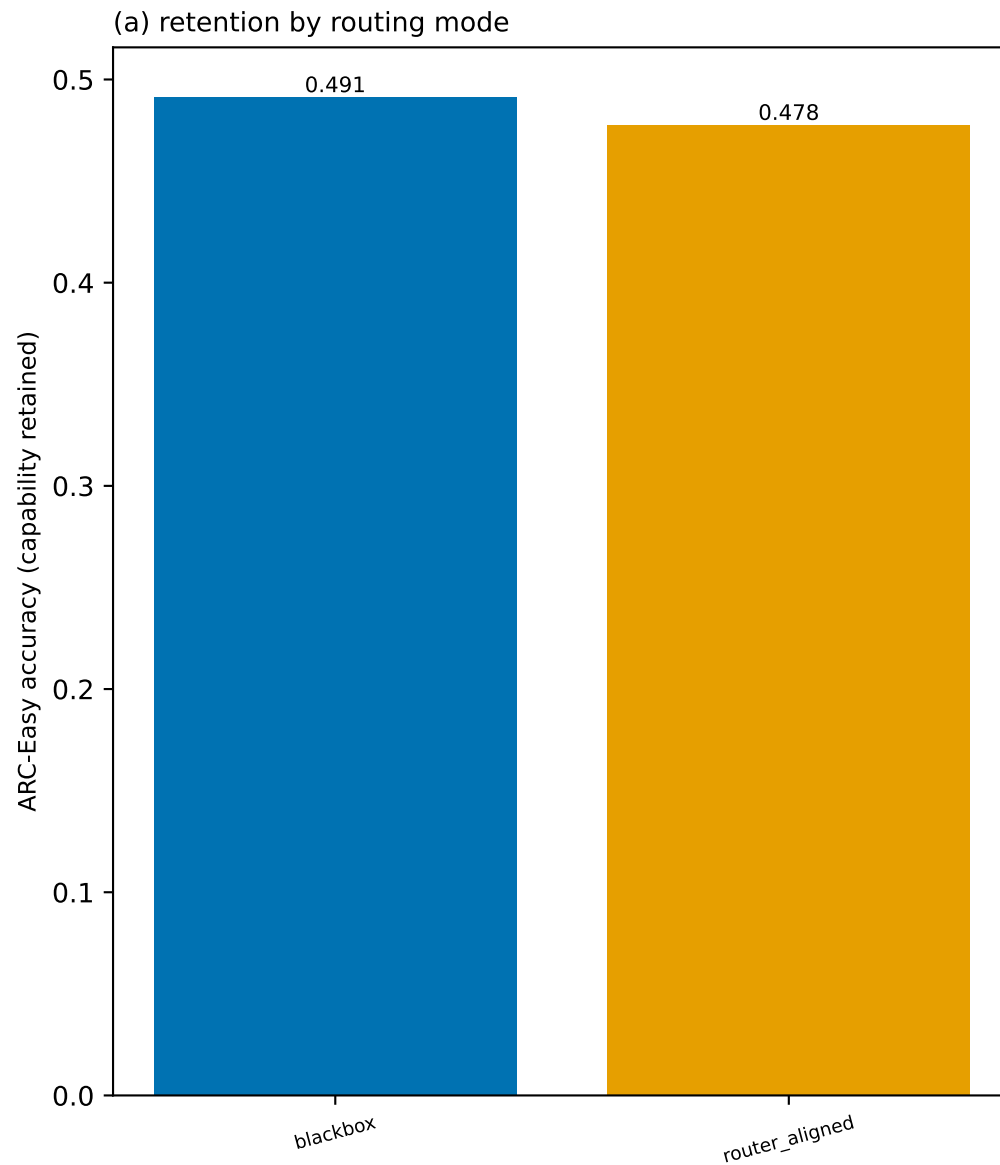
What to memorize — surprise decomposition (0.6B)

Surprise = student-view NLL minus teacher-view NLL on each answer token (base model, privileged block removed vs present). Footprint over content heads, structural/sink tokens masked. High surprise splits into knowledge_gap (604; teacher attends the privileged block) and misdirection (15; teacher attends shared in-context tokens). Knowledge gaps are the memorization target; misdirection is a routing fix.



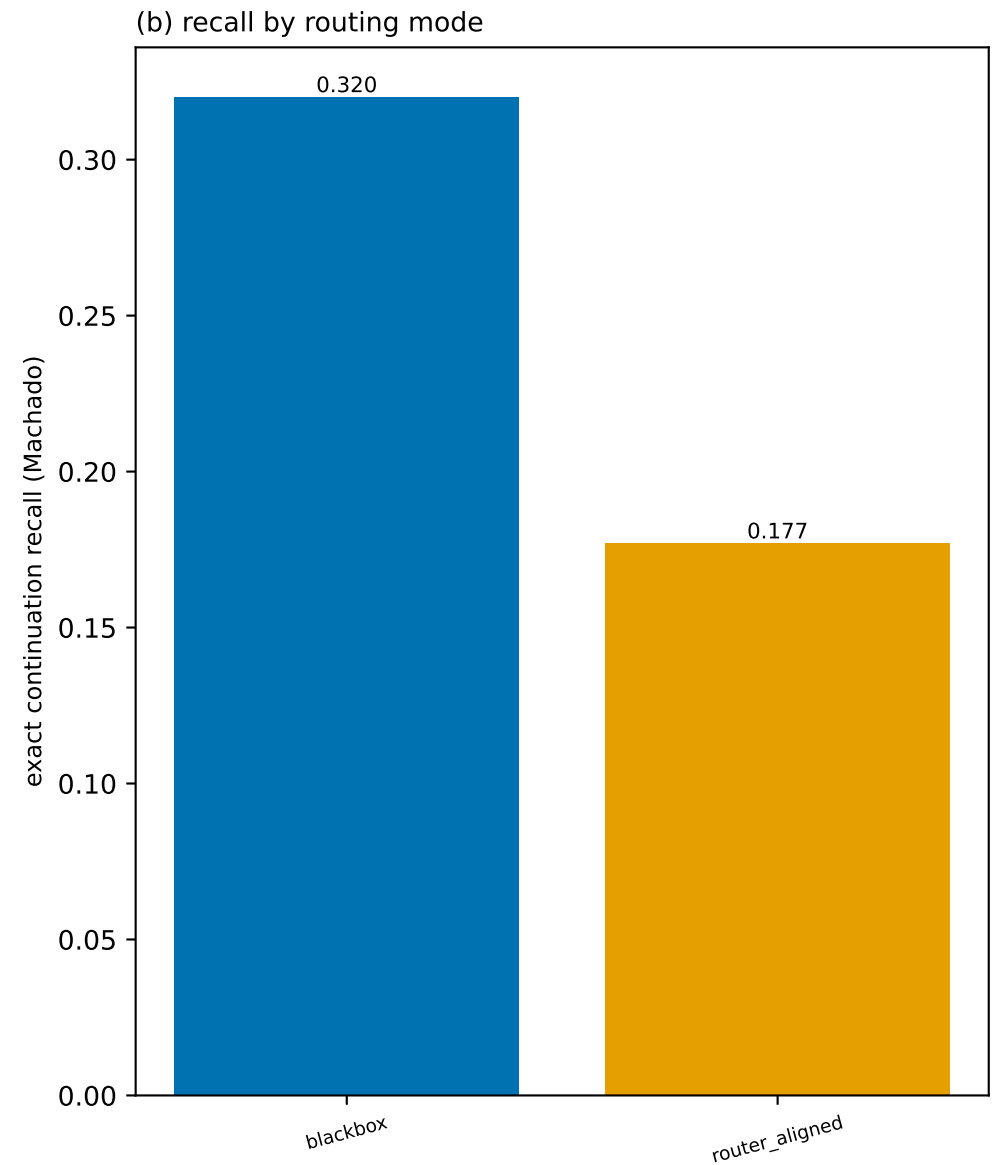
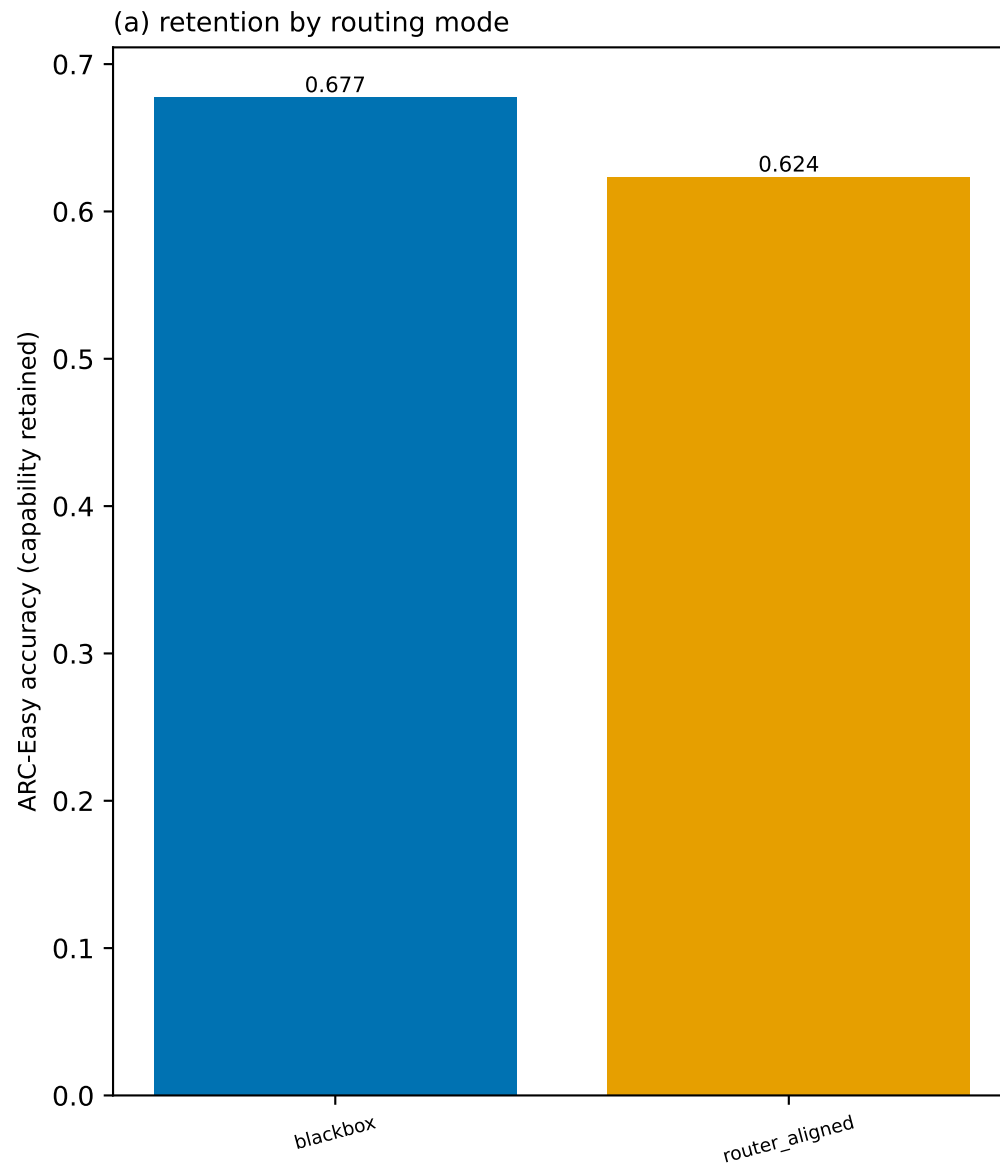
Routing: blackbox vs explicit router — Qwen3-0.6B

Same base, three routing supervisions. If blackbox matches the aligned arms in both axes, the optimistic 'blackbox converges' conjecture holds; if the aligned arms lead on retention or recall, blackbox under-converges on the routing part.



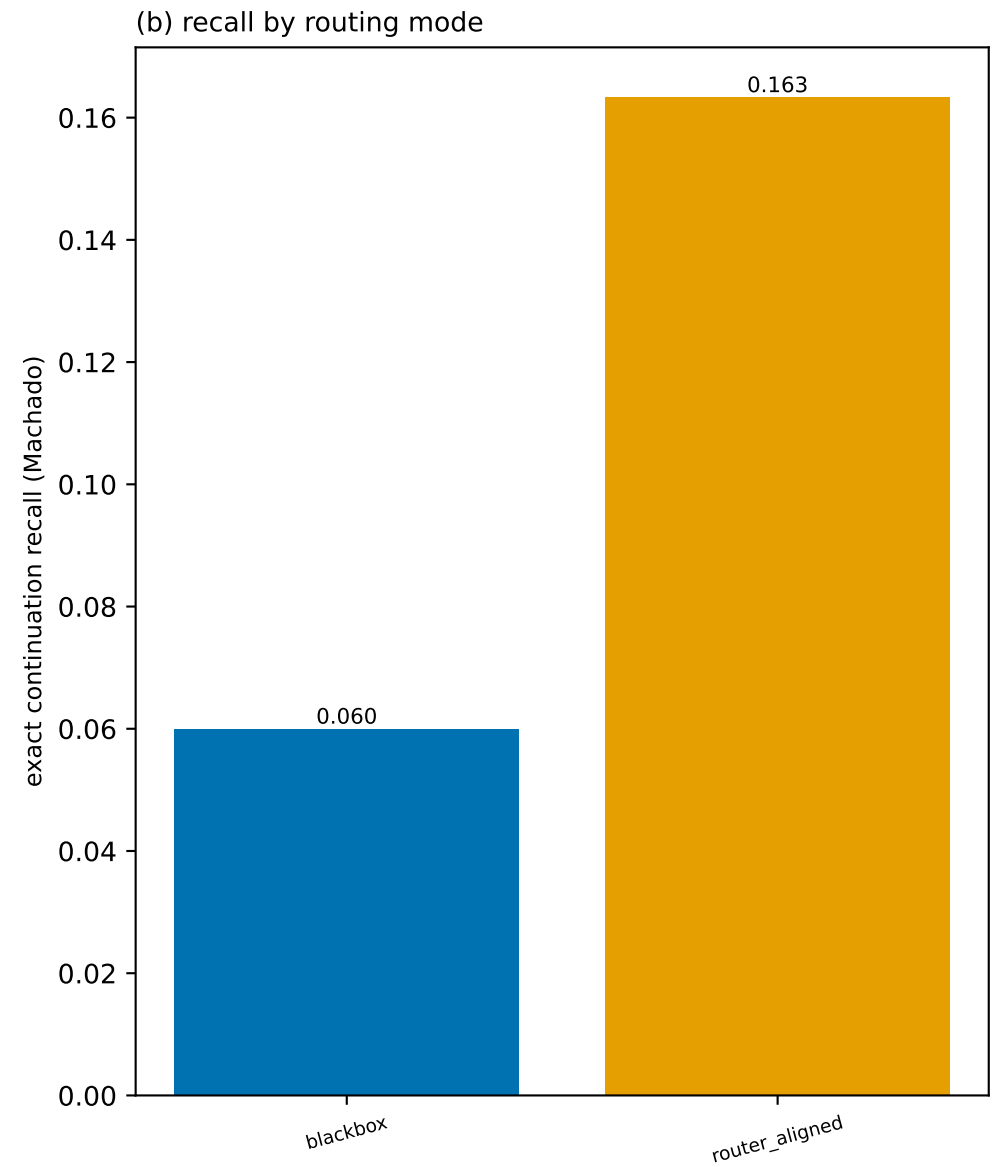
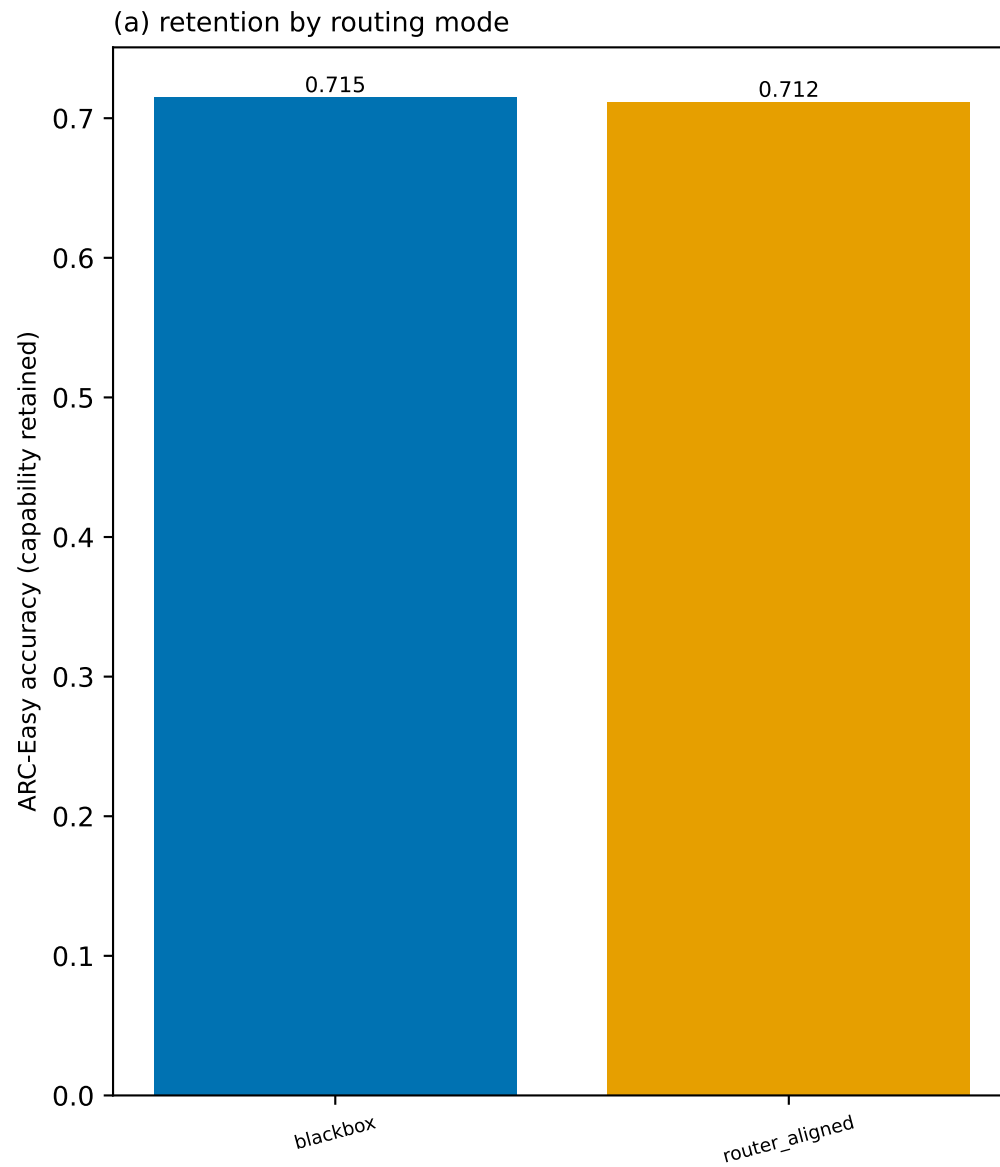
Routing: blackbox vs explicit router — Qwen3-1.7B

Same base, three routing supervisions. If blackbox matches the aligned arms in both axes, the optimistic 'blackbox converges' conjecture holds; if the aligned arms lead on retention or recall, blackbox under-converges on the routing part.



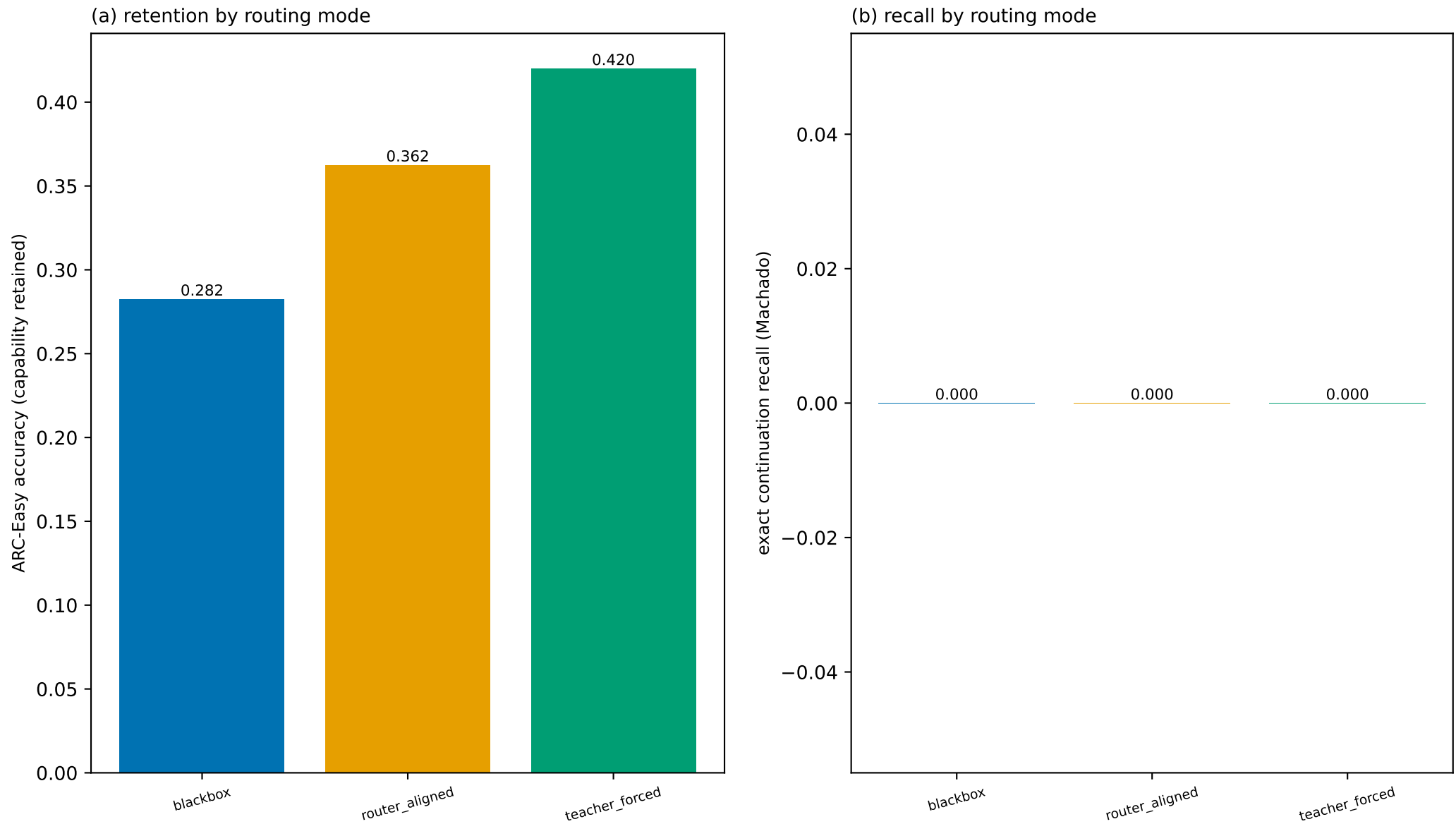
Routing: blackbox vs explicit router — Qwen3-4B

Same base, three routing supervisions. If blackbox matches the aligned arms in both axes, the optimistic 'blackbox converges' conjecture holds; if the aligned arms lead on retention or recall, blackbox under-converges on the routing part.



Routing: blackbox vs explicit router — gpt-oss-20b

Same base, three routing supervisions. If blackbox matches the aligned arms in both axes, the optimistic 'blackbox converges' conjecture holds; if the aligned arms lead on retention or recall, blackbox under-converges on the routing part.



Best retention row per method family and model

Recall: CER (lower better) and exact-match probes (higher better). Retention: ARC retained and WikiText log(ppl / teacher ppl) damage.

source	model	run	recall_cer	cont_mach	cloze_mach	sensor_cerv	arc_acc	arc_retain	wiki_log_damage
layerwise	Qwen3-0.6B	clean_q_ch1_slide8_vocab_0p6b_	0.980	0.000	0.009	0.886	0.573	1.070	0.685
layerwise	Qwen3-1.7B	dev	—	0.320	0.936	0.102	0.677	1.034	-0.029
layerwise	Qwen3-4B	lw_l_tailonly_4b_rag	0.013	0.380	0.982	0.136	0.738	0.987	0.160
classical-kd	Qwen3-0.6B	kd_full_0p6b_rag	0.863	0.410	0.991	0.034	0.555	1.037	0.258
multigpu	gpt-oss-20b	mlt_gptoss20b_slide2_tf_lora	—	0.000	0.000	0.011	0.420	0.568	1.537

Retention coverage — remaining work

Runs WITHOUT the new retention battery (need scripts/retention_eval.py):

```
[layerwise] 25 missing
  clean_combined_slide8_vocab_14b_lora   clean_combined_slide8_vocab_8b_lora   clean_machado_slide8_vocab_14b_lora
  clean_machado_slide8_vocab_8b_lora   clean_machado_slide8_vocab_alia40b_lora   clean_q_ch1_slide2_vocab_14b_lora
  clean_q_ch1_slide4_vocab_14b_lora   clean_q_ch1_slide8_lenskl_14b_lora   clean_q_ch1_slide8_nmse_14b_lora
  clean_q_ch1_slide8_vocab_14b_lora   clean_q_ch1_slide8_vocab_8b_lora   clean_q_ch1_slide8_vocab_alia40b_lora
  lw_j_llama8b_rag   lw_j_mistral7b_rag   lw_j_phi4mini_rag
  lw_j_qwen8b_rag   lw_k_gptoss_rag   lw_l_final_14b_rag
  lw_l_final_8b_av2_rag   lw_l_final_8b_rag   lw_p_ft4b_rag
  lw_tc_lora_14b_rag   smoke_bucketed_mb2_gpu3   smoke_item_mb2_gpu3
  tl_i_tunedlenskl_strict_0p6b_rag

[classical-kd] 3 missing
  kd_lora_kl_hi_e40_v3_qwen36_27b_rag   kd_lora_kl_hi_e40_v3_qwen3_30ba3b_inst2507_rag   kd_lora_kl_hi_e60_v3_14b_rag

[multigpu] 14 missing
  mlt_alia40b_slide2_lora   mlt_gemma4_26ba4b_slide2_lora   mlt_gemma4_26ba4b_slide2_ra_lora
  mlt_gemma4_26ba4b_slide2_tf_lora   mlt_gemma4_31b_kllocal_lora   mlt_gemma4_31b_slide2_lora
  mlt_gemma4_31b_slide3_lora   mlt_gptoss120b_auto4_slide2_tf_lora   mlt_gptoss120b_pp2_kllocal_lora
  mlt_gptoss120b_pp2_slide2_lora   mlt_gptoss120b_pp2_slide2_tf_lora   mlt_gptoss120b_pp2_slide3_lora
  mlt_qwen35_122b_auto4_slide2_lora   mlt_qwen35_122b_gptq_auto2_slide2_lora
```